

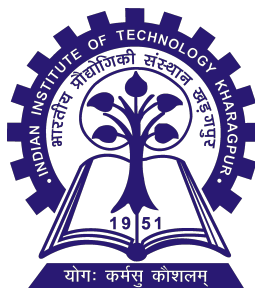
Master's Thesis Project Report
Academic Year: 2025 - 2026

**From Molecules to Qubits: Unravelling Noise in
Self-Assembled Monolayers**

Thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science (Chemistry)

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April 2026

DEDICATION

To the children whose stories ended before they began,
this work honours the lives war could not.

May their memory forever call us toward peace and compassion

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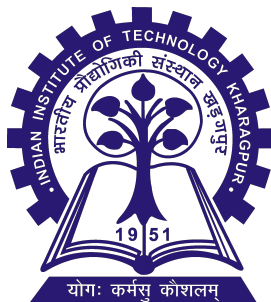
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Certificate

This is to certify that **Mr. Arif Ahmed Begg**, a second-year M.Sc. student of the Department of Chemistry, Indian Institute of Technology Kharagpur, bearing roll number **24CY40005**, has successfully completed his master's thesis project under my supervision during the period July 2025 to April 2026. During this time, he has worked on a project entitled "**From Molecules to Qubits: Unravelling Noise in Self-Assembled Monolayers**".

He demonstrated a strong work ethic, a willingness to learn, and contributed effectively to the tasks assigned. I rate his performance during the project as **excellent**. The undersigned may be contacted for any further information regarding the student.

Prof. Sabyashachi Mishra
Professor
Department of Chemistry, IIT Kharagpur

Place: IIT Kharagpur

Declaration

I, Arif Ahmed Begg, hereby certify that:

1. The work contained in this thesis is original and has been carried out by me under the supervision of my supervisor and guide.
2. This work has not been submitted to any other institute for the award of any other degree.
3. I have followed the guidelines provided by the institute in the preparation and writing of this thesis.
4. I have adhered to the norms and guidelines outlined in the Ethical Code of Conduct of the Institute.
5. Wherever I have used material (data, theoretical analysis, or text) from other sources, I have duly acknowledged them by citing them in the text and listing them in the references.

Place: IIT Kharagpur

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Date:

Abstract

A clear understanding of noise and decoherence is crucial for making molecular qubits more reliable. This work investigates the microscopic origin of noise and decoherence in self-assembled monolayer (SAM) molecular qubits, with a focus on radical-based spin systems. While advances in quantum hardware have enabled significant scaling of qubits, maintaining coherence and minimizing environmental noise remain critical challenges. In SAM-based architectures, qubit performance is strongly influenced by intermolecular interactions, substrate effects, and vibrational dynamics. Using a combination of normal mode analysis, Wigner sampling, and spin Hamiltonian modeling, we examine how structural distortions affect anisotropy through variations in the g- and A-tensors. The study further distinguishes between edge and central radicals in SAM (having different environment), revealing differences in spin-spin coupling and dynamics. Our results show that local molecular environment and collective motion play a key role in determining decoherence pathways.

1 Introduction

A qubit is the fundamental unit of quantum information that encodes data using quantum mechanical phenomena such as superposition and quantum entanglement. Unlike a classical bit that is restricted to the binary states 0 or 1, a qubit can exist in a coherent linear combination of both states simultaneously and can be expressed as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex probability amplitudes satisfying the normalization condition. This quantum property enables quantum processors to explore exponentially large computational state spaces with a relatively small number of qubits.

The evolution of quantum computing hardware reflects a dual focus on increasing qubit count and enhancing operational fidelity [2, 3]. Early superconducting processors contained a few dozen qubits, which enabled initial demonstrations of quantum circuits but were limited by noise and short coherence times. Subsequent generations scaled to over one hundred and then several hundred qubits, marking significant progress in device fabrication and control. However, simply increasing qubit number is insufficient; error rates, connectivity, and coherence must also improve [4]. Modern roadmaps therefore emphasize error mitigation, and advanced runtime software to stabilize computations.

The application of a practical quantum information processor requires meeting the five criteria proposed by DiVincenzo [5, 6, 7]. First, the system must consist of a scalable set of well-characterized qubits with known internal Hamiltonian parameters. It should also allow reliable initialization of the qubits into a simple state such as $|0\rangle$. The coherence time of the qubits must be much longer than the time required for gate operations to reduce the effects

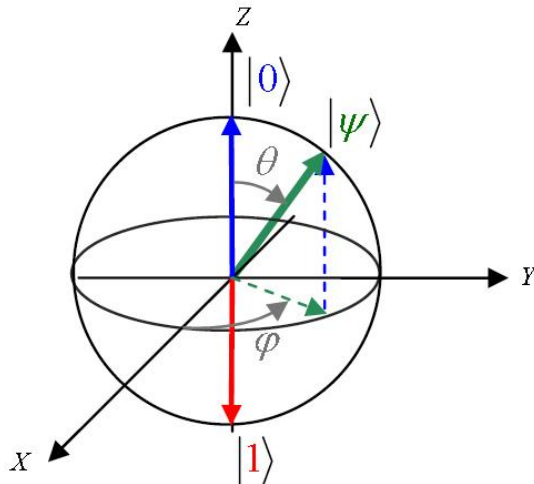


Figure 1: Qubit initialization schematics. $|\Psi\rangle$ is the superposition state of $|0\rangle$ and $|1\rangle$. θ : angle from the $+Z$ axis; ϕ : angle in the XY -plane from $+X$ axis. The image is reproduced from math.stackexchange.com.

of decoherence. In addition, the system must support a universal set of quantum gates to perform arbitrary quantum operations. Finally, it must allow qubit-specific measurements to read out the final state of the computation.

Quantum computing runs quantum algorithms capable of solving problems that are practically unsolvable for classical computers. The implementation of quantum algorithms can be decomposed into a set of single-qubit and two-qubit universal quantum logic gates [8, 9, 10, 11]. Over the past decade, considerable progress has been made across various implementations, including superconducting circuits, trapped ions, semiconductor spin qubits, molecular qubits, and photonic qubits [11, 12, 13]. Although many candidates demonstrate excellent performance at the single-unit level, integrating multiple qubits to enable quantum gate operations remains difficult [14, 15, 16]. Organic polymers could not integrate radicals as monomers [17, 18]. The spatial distribution of radicals could not be designed by side-chain engineering or block copolymer self-assembly. Establishing controlled coupling between several qubits is technically challenging, and large-scale implementation is further constrained by the lack of fabrication methods capable of precisely positioning qubits in ordered arrays [18, 19, 20, 21]. These requirements are essential for constructing functional quantum devices.

Molecular qubits have emerged as a promising approach to overcome these limitations. Molecules behave as microscopic quantum systems, similar to atoms but with greater structural flexibility. This versatility allows them to form well-organized nanoscale architecture. Moreover, their synthetic tunability enables precise control over interactions between the

qubits. By modifying the surrounding ligand environment, molecular qubits can also be adapted for specific applications, such as integration onto solid substrates or incorporation into functional quantum devices [13].

The quantum behaviour of molecular qubits can be rigorously described using an effective spin Hamiltonian, which accounts for the interaction of the spin with an external magnetic field and with nearby nuclear spins:

$$\hat{H} = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \hat{\mathbf{S}} + \hat{\mathbf{S}} \cdot \mathbf{A} \cdot \hat{\mathbf{I}} \quad (1)$$

where μ_B is the Bohr magneton, $\mathbf{B}$ is the external magnetic field, $\hat{\mathbf{S}}$ and $\hat{\mathbf{I}}$ are the electron and nuclear spin operators, respectively. The $\mathbf{g}$ -tensor describes the anisotropic Zeeman interaction, and the hyperfine tensor $\mathbf{A}$ represents the coupling between the electron spin and nearby nuclear spins.

Despite these advances, large-scale quantum computation remains limited by decoherence, operational noise, and scalability constraints. Integrating a large number of qubits into a single scalable architecture remains a major engineering challenge. Current quantum processors require complex control electronics, cryogenic environments, and precise calibration of quantum gates. As the number of qubits increases, the signal-to-noise ratio decreases, and control circuitry becomes increasingly complex. Furthermore, maintaining uniform qubit performance across large-scale integrated systems is difficult due to variations in materials, material defects, and thermal fluctuations.

Noise in simple qubit systems primarily arises from interactions between the qubit and its surrounding environment [22, 23, 24]. In solid-state implementations such as superconducting and semiconductor qubits, common noise sources include charge fluctuations and electromagnetic disturbances [25, 26]. Charge fluctuations arise from randomly trapped charges in device materials or interfaces, leading to fluctuations in the qubit transition frequency and dephasing. Magnetic flux noise is typically generated by fluctuating spins or magnetic defects on material surfaces [27, 28, 29].

In self-assembled monolayer (SAM) radical qubits, as in TEMPO-SAM, noise mainly arises from interactions between molecular spins and their environment [1, 30]. The primary sources include spin fluctuations, dipole-dipole interactions between nearby molecules, and surface-induced disturbances from the substrate. Additional contributions may arise from thermal and vibrational fluctuations of the substrate. Nevertheless, experiments show that properly engineered SAM structures can maintain relatively long spin relaxation times despite surface integration.

Spin-spin interactions are expected to reduce the coherence time of the qubits in SAM.

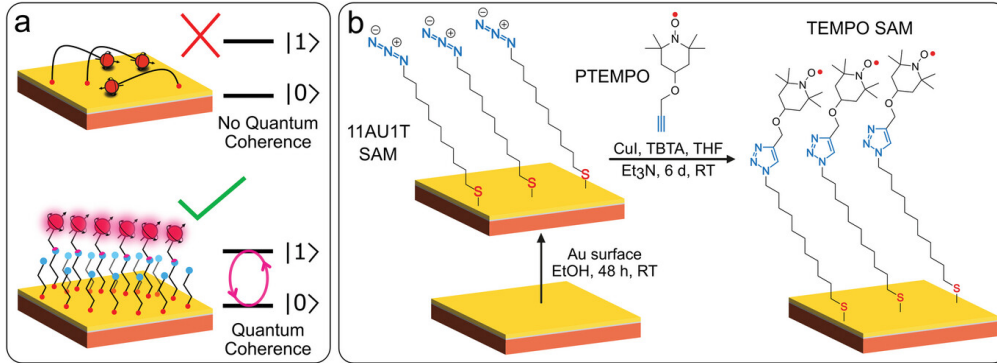


Figure 2: TEMPO-SAM: (a) Ordered TEMPO qubit array scheme (b) Two-step TEMPO assembly on Au surface. The figure is reproduced from reference [1].

Interestingly, the measured phase memory time, T_m values for the TEMPO SAM is approximately three to four times larger than those observed for a single P-TEMPO radical, even when the latter is prepared in a diluted solution [1]. This behaviour may be attributed to strong spin-exchange interactions, which can suppress decoherence effects, as previously reported for spin chain systems. Another possible explanation relates to the two-dimensional structure of the TEMPO SAM, which limits the extent of the nuclear spin bath that contributes to electron spin decoherence [31]. However, a detailed quantum-mechanical investigation of the origin of noise, spin-dynamics and the associated decoherence pathways in SAM-type systems has not yet been reported. Addressing this gap forms the primary goal of the present study.

2 Objectives

- Identify the origin of noise in SAM molecular qubits by analyzing interactions between molecular spins, substrate effects, and environmental fluctuations.
- Investigate the role of molecular vibrations and structural distortions, particularly along the N–O bond, in influencing spin properties such as the g -tensor and A -tensor.
- Comparison of the spin dynamics of radicals within the SAM and understanding how their local environments affect spin-spin coupling and coherence.
- Establish the relationship between molecular arrangement, collective motion, and decoherence pathways, providing information on stability and performance.

3 Computational Methodology

A multiscale computational framework was employed that combines quantum mechanical calculations with classical molecular dynamics (MD) simulations to investigate the structural and dynamical behaviour of TEMPO-based SAMs. This approach was necessary to balance the need for electronic-level accuracy with the ability to access longer timescales in large systems.

3.1 Electronic Structure Calculations of Radicals

DFT Calculations of Spin-Parameters All quantum chemical calculations were performed using the ORCA program package. Geometry optimizations were carried out within the framework of density functional theory (DFT) using the UB3LYP functional with the def2-TZVP basis set. To assess the dependence of g and A -tensors properties on the choice of functional, a functional benchmarking study was conducted using B3LYP, B2PLYP, DSD-PBEP86, and RI-MP2 methods, using the EPR-II basis set, which is specifically designed for magnetic property calculations. The optimized geometries obtained at the DFT level served as the starting point for subsequent vibrational and dynamical analyses. The spin-phonon coupling parameters were evaluated by distorting the molecule along its normal modes.

Vibrational Sampling and Ab Initio Dynamics Thermal fluctuations were incorporated through Wigner sampling based on the harmonic vibrational modes of the optimized structures. This procedure generates an ensemble of initial geometries that reflect zero-point motion and finite-temperature effects, thereby providing a more realistic representation of the system compared to a single optimized configuration. Ab initio molecular dynamics (AIMD) simulations were then performed at 300 K within the NVT ensemble formalism using a Berendsen thermostat. The same level of theory (uB3LYP/def2-TZVP) was retained to ensure consistency with the electronic structure calculations. These simulations provided time-dependent structural trajectories, which were subsequently used to analyze the coupling between molecular vibrations and spin degrees of freedom.

3.2 Molecular Dynamics Simulations of Self-Assembled Monolayer

Modeling Self-Assembled Monolayer Given the large size of the SAM system ($> 1.25 \times 10^4$ atoms), classical MD simulations were employed to access longer timescales beyond the reach of ab initio methods. A full three-dimensional model of the TEMPO SAM was constructed to retain intermolecular interactions that are absent in simplified

two-dimensional representations. Initial coordinates were generated using in-house Python scripts. An Au(111) surface was constructed using VESTA, and alkanethiolate TEMPO molecules were arranged to reproduce experimentally relevant surface coverage. The final model consisted of 141 molecules distributed over an area of 3525 Å², corresponding to a surface density consistent with experimental estimates. K-means clustering was used to identify adsorption sites and ensure a uniform spatial distribution of molecules.

Force Field and Simulation Description Classical MD simulations were carried out using a combination of an interfacial force field for the gold substrate and the General Amber Force Field (GAFF) for the organic components (C, H, N, O and S). The system was first energy-minimized for 10,000 steps to remove unfavourable contacts, followed by gradual heating from 0 to 300 K over a period of 1 ns. Production simulations were performed for 10 ns at 300 K within the NVT ensemble using a Langevin thermostat. Additional simulations at 100 K were conducted for comparison. A timestep of 1 fs was used throughout. To preserve the structural integrity of the substrate, Au and S atoms were restrained during the simulations. All simulations were performed under vacuum conditions to isolate intrinsic interfacial effects. Analyses such as RMSD were checked using the VMD software.

4 Results and Discussion

4.1 Spin-Relaxation in TEMPO and P-TEMPO Radicals

4.1.1 DFT Functional Benchmarking

A comparative assessment of different functionals was carried out in order to identify a suitable level of theory for accurately describing the electronic and spin properties of the TEMPO radical. For the hyperfine coupling constant, B3LYP shows a positive deviation, indicating an overestimation relative to the theoretical value of $g_{\text{iso}} = 2.0063$. (The isotropic g -value is defined as $g_{\text{iso}} = \frac{1}{3}(g_x + g_y + g_z)$, i.e., the trace average of the g -tensor. It represents the orientation-averaged Zeeman interaction of the spin, typically observed under fast-tumbling conditions). In contrast, B2PLYP yields a small negative deviation, significantly closer to the reference value compared to the other methods. DSD-PBEP86 and RI-MP2 both underestimate A_{iso} more strongly, with RI-MP2 showing the largest deviation among all methods considered. This suggests that while higher-level correlation methods improve certain aspects of the electronic description, they may overcorrect for hyperfine interactions. A similar trend is observed for the g_{iso} values. B3LYP underestimates the g -factor, while RI-MP2 slightly overestimates it. B2PLYP and DSD-PBEP86 both provide values closer to the ref-

erence, with B2PLYP showing particularly good agreement. Notably, B2PLYP achieves a balanced description of both A_{iso} and g_{iso} , whereas other methods tend to perform well for one property at the expense of the other. Based on this analysis, B2PLYP was selected as

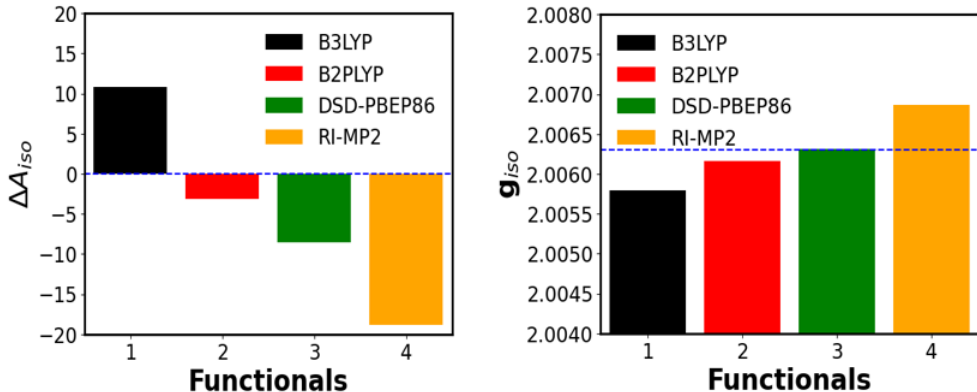


Figure 3: Benchmarking of DFT functionals for TEMPO involving B3LYP, B2PLYP and DSD-PBEP86, and RI-MP2 with EPR-II basis set. B2PLYP method shows the best agreement with the theoretical values, which was used for further calculations.

the preferred functional for subsequent calculations, as it gives the most balanced agreement for both hyperfine and g -values, whereas other methods either overestimate or underestimate one of them significantly (Refer to Figure 3).

4.1.2 Spin Hamiltonian Parameters

The electronic structure of TEMPO provides well-defined spin states that can be manipulated using external magnetic fields. The spin distribution shows that the unpaired electron is predominantly localized on the nitroxide moiety, with only minor delocalization over the surrounding atoms.

4.1.3 Normal Modes of Vibration

To investigate the origin of the anisotropy in the g and A tensors, the TEMPO and P-TEMPO molecules were distorted along its normal vibrational modes. For a molecule containing N atoms, the vibrational degrees of freedom are described by its $3N - 6$ normal modes. Each of these modes represents an independent collective nuclear motion that can potentially modulate the electronic structure of the radical centre and thereby influence the magnetic anisotropy parameters.

However, in practice, not all vibrational modes are thermally populated at finite temperature. Only those modes whose energies are comparable to or lower than the available

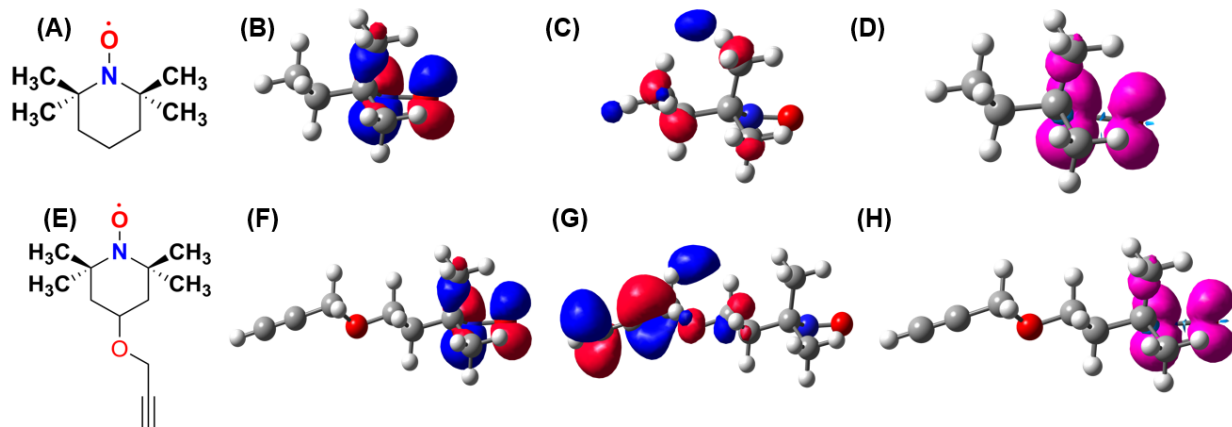


Figure 4: Representation of (A): Structure (B): HOMO (C): LUMO (D): Spin density of TEMPO; Representation of (E): Structure (F): HOMO (G): LUMO (H): Spin density of P-TEMPO. Spin density primarily localized along the N–O bond.

thermal energy are expected to contribute significantly to structural fluctuations. Within the harmonic oscillator approximation, the thermal energy available at temperature T is approximately $k_B T$, which provides an estimate for the maximum vibrational frequency that can be appreciably populated at temperature T . This analysis predicts a cutoff frequency of $\approx 208.5 \text{ cm}^{-1}$ at 300 K. Consequently, only the lowest-energy vibrational modes ($\leq$ cutoff frequency) are expected to play an important role in modulating the magnetic parameters at low temperatures.

From a quantum mechanical perspective, the *ab initio* electronic structure calculations performed in this work correspond to the zero-temperature limit, where nuclear motion is treated within the harmonic approximation around the equilibrium geometry. To account for the influence of nuclear vibrations on the magnetic tensors, the molecule was systematically displaced along each normal mode coordinate Q_k . By evaluating the magnetic parameters for positive and negative distortions along these coordinates, the sensitivity of the g and A tensors to vibrational motion can be quantified. To complement this deterministic distortion approach, Wigner sampling was employed to generate nuclear configurations that incorporate quantum-mechanical zero-point motion and vibrational fluctuations around the equilibrium structure.

The combined analysis of systematic normal-mode distortions and Wigner-sampled configurations allows us to identify the vibrational motions that most strongly influence the magnetic anisotropy. Our preliminary results indicate that only those vibrational modes capable of significantly perturbing the N–O bond geometry contribute appreciably to the variation of the g and A tensors. In particular, the dominant contributions arise from vibrational motions involving N–O stretching, O–N–R in-plane bending, nitrogen pyramidalization, and

out-of-plane dihedral distortions. These modes effectively modulate the electronic structure of the nitroxide radical center and therefore play a key role in determining the magnitude of the spin-phonon coupling and the resulting anisotropy parameters.

A number of previous studies have considered modified systems such as P-TEMPO, in which a propargyl group is attached to the TEMPO framework. Apart from its potential application in forming self-assembled monolayers through click reactions, P-TEMPO possesses a larger number of vibrational modes. Consequently, it is expected that a greater number of vibrational modes may influence the N–O bond, which could allow the system to remain active at higher temperatures compared to TEMPO, and is the subject of our study. Performing the same exercise as in TEMPO, the first eight modes were considered active at room temperature.

Table 1: Normal modes of vibration, their frequency, type, and normalised vibrational mode contributions for P-TEMPO (first 8 modes).

#Mode	Frequency (cm ⁻¹)	Type	N–O (%)	Ring (%)	Tail (%)
1	29.8	Wagging	0.7	14.4	84.9
2	56.7	Wagging	6.8	54.9	38.3
3	72.8	Twisting	0.8	35.0	64.2
4	86.6	Twisting	4.0	77.3	18.7
5	116.4	Twisting	0.0	96.2	3.8
6	158.8	Twisting	0.4	42.4	57.2
7	196.7	Twisting	5.5	94.2	0.3
8	207.4	Methyl rotation	0.0	99.9	0.1

4.1.4 Spin-Vibration Coupling

To quantify the dependence of the magnetic parameters on molecular distortion, the variation of the property of interest (g -tensor and A -tensor) with respect to the selected normal mode coordinates was analyzed using a Taylor series expansion. In this framework, the zeroth-order term corresponds to the equilibrium value of the property and represents the Zeeman contribution, which remains constant for a given molecule at its equilibrium geometry. For a general property P that depends on the normal mode coordinates Q_k , the Taylor series expansion can be expressed as [32].

$$P(\{Q_k\}) = P^{(0)} + \sum_k \left(\frac{\partial P}{\partial Q_k} \right)_0 Q_k + \frac{1}{2} \sum_{k,l} \left(\frac{\partial^2 P}{\partial Q_k \partial Q_l} \right)_0 Q_k Q_l + \dots \quad (2)$$

For small enough distortion, truncation up to the second term is sufficient. The first-order term describes the linear response of the property to vibrational distortion, whereas

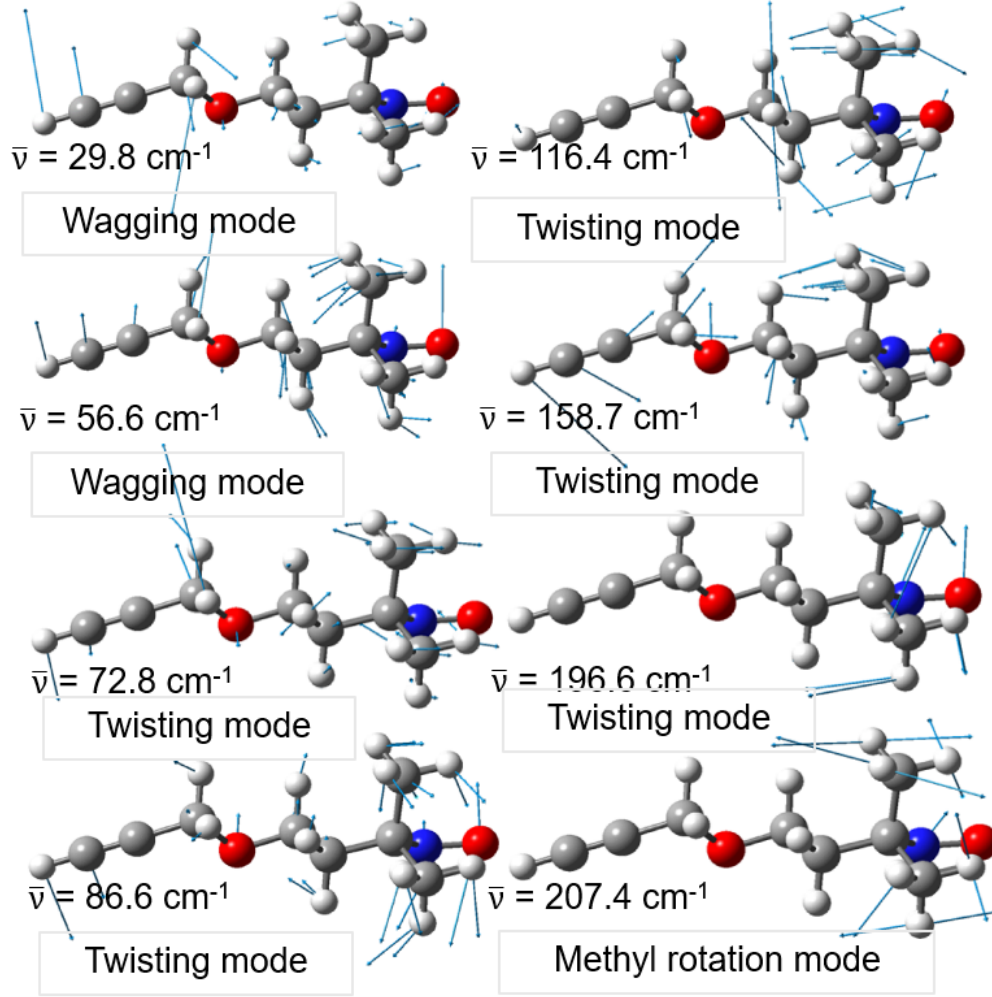


Figure 5: First eight modes of P-TEMPO. Modes classified to wagging, twisting and methyl rotation.

the second-order term accounts for quadratic contributions. A linear dependence of A on Q indicates that the dominant contribution arises from vibration-induced perturbations of the nuclear spin density. In contrast, a quadratic dependence of A on Q suggests that dipole-dipole interactions also contribute to the coupling. A net spin-phonon coupling for the selected modes was evaluated using the following expression:

$$\Lambda_k^{(\text{net})} = \frac{1}{2\hbar\omega_k} \left[\left(\frac{\partial g}{\partial Q_k} \right)^2 + \left(\frac{\partial A}{\partial Q_k} \right)^2 \right] \quad (3)$$

where the first-order couplings (Zeeman terms) are only considered. This gives an idea among the modes active at room temperature, which should affect the most in bringing the spin-phonon coupling. Clearly the net spin-phonon coupling is more for P-TEMPO compared to

that of TEMPO, except mode 3.

Further, the spin-phonon coupling in the modes has been classified into three segments, the contribution from the ring (piperidine and the associated methyl groups), tail (alkyl chain) and the N-O bond vector. Modes 2, 4, and 7 of P-TEMPO show the maximum contribution arising from N-O, which are significantly better contributors to anisotropies.

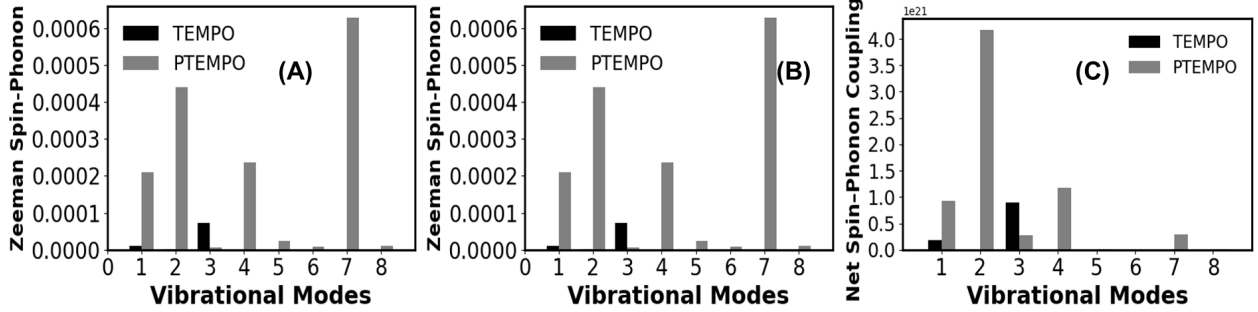


Figure 6: Zeeman spin-phonon and hyperfine splitting for TEMPO and P-TEMPO.

4.1.5 Spin-Lattice Relaxation Time

Spin relaxation arises primarily from two mechanisms: spin-spin interactions and spin-lattice coupling. Spin-spin interactions are generally temperature independent and occur through flip-flop processes between nearby electron and nuclear spins. In contrast, spin-lattice coupling is temperature dependent and results from interactions between the spin system and lattice vibrations. This mechanism includes several processes such as the direct, Raman, Orbach, local-mode, thermally activated, and tumbling-dependent processes. Spin decoherence, on the other hand, is mainly caused by environmental magnetic noise. The principal physical mechanisms responsible for this effect include instantaneous diffusion and spectral diffusion, both of which lead to the loss of phase coherence of the spin states. The spin-lattice relaxation time T_1 characterizes the rate at which a spin system returns to thermal equilibrium with the lattice after excitation. The relaxation occurs through interactions between the electronic spins and lattice vibrations (phonons). The total relaxation rate is generally expressed as a sum of different phonon-mediated processes:

$$\frac{1}{T_1} = \left(\frac{1}{T_1}\right)_{\text{Direct}} + \left(\frac{1}{T_1}\right)_{\text{Raman}} + \left(\frac{1}{T_1}\right)_{\text{Orbach}} \quad (4)$$

Each relaxation mechanism dominates in a different temperature regime. The direct process is a one-phonon mechanism in which the spin transition occurs through the absorption or emission of a single phonon whose energy matches the Zeeman splitting of the spin levels.

From the frequency analysis, the fitting was performed considering only the direct and Raman processes in the temperature range 7-60 K, via [33, 34, 35].

$$\frac{1}{T_1} = A \frac{\exp\left(\frac{\hbar\omega_1}{k_B T}\right)}{\exp\left(\frac{\hbar\omega_1}{k_B T}\right) - 1} + B \sum_i V_i \frac{\exp\left(\frac{\hbar\omega_i}{k_B T}\right)}{\left(\exp\left(\frac{\hbar\omega_i}{k_B T}\right) - 1\right)^2} + C \quad (5)$$

Here, T_1 denotes the spin-lattice relaxation time; A , B , and C represent the strengths of the direct, residual, and phonon-mediated relaxation processes, respectively; ω_1 is the Zeeman angular frequency, ω_i are the normal mode frequencies, V_i are the corresponding spin-phonon coupling coefficients, and $\hbar$ and k_B are the reduced Planck and Boltzmann constants.

The first term represents the direct one-phonon process, where A is the coupling constant and ω_i is the frequency of that mode. The second term B represents a temperature independent background relaxation. The third term corresponds to the multi-phonon process involving vibrational modes of frequency ω_i , where C is the coupling constant, and V_i represents the strength of the spin-phonon coupling for each mode. The total relaxation rate is composed of a direct one-phonon process, a temperature-independent residual contribution, and a phonon-assisted (multi-mode) process governed by spin-vibrational coupling.

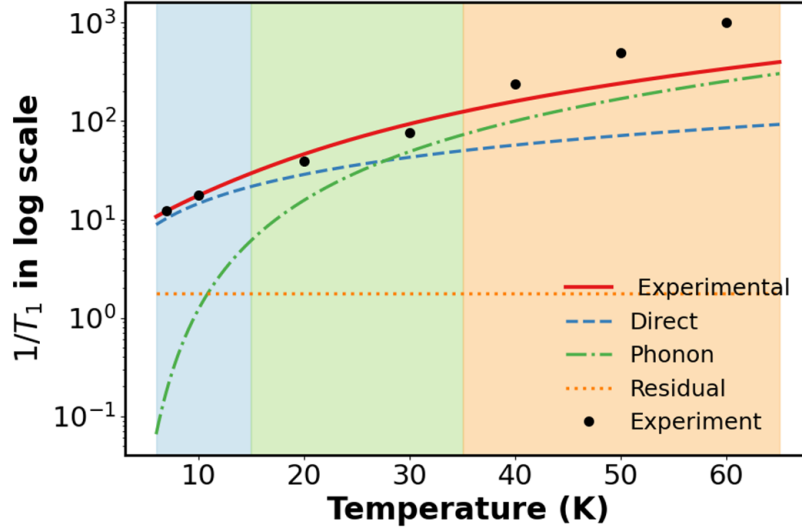


Figure 7: T_1 relaxation fit (Equation 5) along with the % contribution of different relaxation mechanisms in different temperature ranges.

Clearly the direct mechanism is prominent at lower temperature range, whereas the Ra-

man processes are well sound at intermediate temperature, owing to the availability of a large number of virtual states.

4.2 Spin-Relaxation in Self-Assembled Monolayer (SAM)

4.2.1 Simulation of Self-Assembled Monolayer

The computational cost of QM calculations increases rapidly with system size, typically scaling between N^3 and N^7 depending on the level of theory. A realistic TEMPO-SAM system contains hundreds to thousands of atoms, including the surface, alkyl chains, and radical centres. In addition, the system involves long-range electrostatic interactions, which further increase the computational complexity. As a result, performing full geometry optimization or AIMD calculations for such systems is computationally impractical. Another important limitation is the accessible simulation timescale. Quantum molecular dynamics methods, such as AIMD or Born–Oppenheimer molecular dynamics (BOMD), require very small time steps of approximately 0.5–1 fs and typically allow total simulation times of only 10–100 ps. However, physical processes in TEMPO-SAM, such as the rotational correlation time of the mobile spin centres, occur on a much longer timescale of approximately 2 ns. Therefore, QM-based molecular dynamics simulations can not directly capture the relevant molecular motions in the system. MD simulations using the Interfacial Force Field (IFF) for Au and the General Amber Force Field (GAFF) for adsorbates at 300 K identified adsorption sites via *k*-means clustering, with TEMPO initially normal to the surface. Systems were equilibrated and run for 10 ns in the canonical ensemble using a Langevin thermostat with a 1 fs timestep under vacuum, with Au and S atoms restrained. Additional simulations at 100 K was performed to study temperature effects.

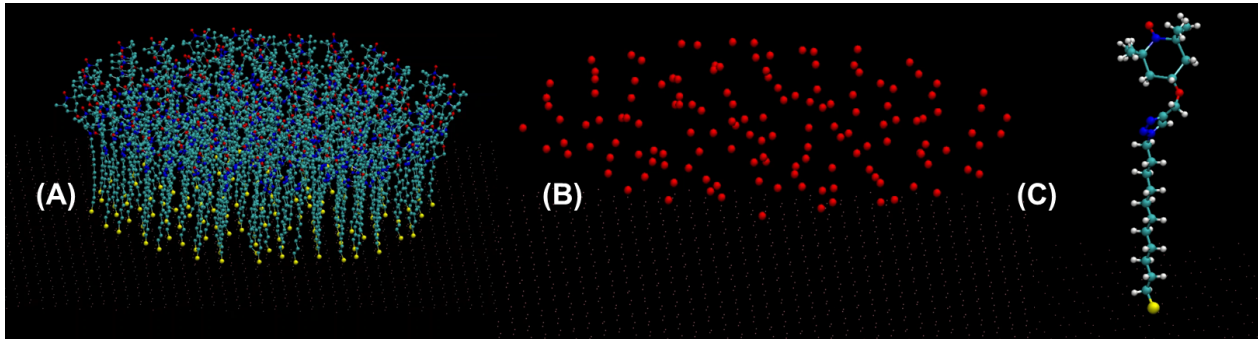


Figure 8: TEMPO-SAM: (A): TEMPO SAM preparation (B): Only the oxygen (radical centre) is highlighted (C): A single radical is shown.

Simulations reveal the presence of two distinct types of radical centres in the system, namely central radicals and edge radicals, which differ in their local environments. The cen-

tral radicals are surrounded by a larger number of neighbouring radicals compared to the edge radicals. This difference is clearly reflected in the radial distribution function $g(r)$ analysis. At 300 K, the first coordination shell for the central radicals appears at approximately 3.25 Å with a pronounced peak, indicating a relatively dense local packing of neighbouring radicals. In contrast, for the edge radicals the first shell occurs at a larger distance of approximately 6 Å, reflecting a more open local environment.

Upon lowering the temperature to 100 K, the intensity of the first-shell peak for the edge radicals decreases, suggesting a change in the local structural organization. Despite the crowding of spin centres around the central radicals, their relative positions exhibit limited fluctuations, indicating a largely static configuration. Such static arrangements contribute significantly to spin coherence, as the system tends to remain in energetically favourable configurations. In contrast, the edge radicals exhibit greater positional fluctuations, leading to a more dynamic local spin environment.

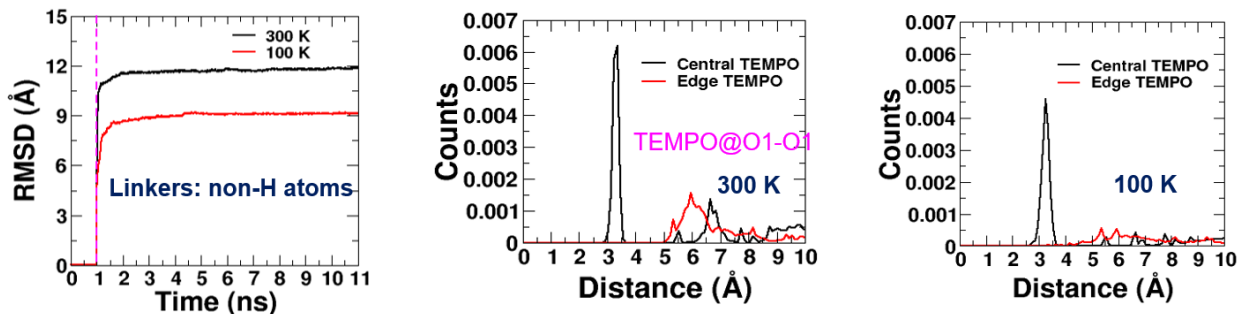


Figure 9: TEMPO-SAM: (A): RMSD analysis at 300 K and 100 K (B): $g(r)$ at 300 K (C): $g(r)$ at 100 K. Fluctuations increase with temperature. Central radicals are better shielded, edge radicals show more dynamics.

The local crowding environment experienced by molecules in the system at a given time can be expressed as a weighted spatial crowding index (η_{CI}) as

$$\eta_{CI}^{(i)}(t) = \sum_{j \neq i} \exp(-\lambda r_{ij}(t)) \quad (6)$$

where $\eta_{CI}^{(i)}(t)$ represents the crowding index of molecule i at time t , and the summation runs over all other radicals j , with $r_{ij}(t)$ as the instantaneous distance between i and j at time t . λ is a decay parameter controlling how rapidly the influence of neighbouring molecules diminishes with distance, defined based on the first-shell separation and distinguished for central and edge radicals ($g(r)$ analysis). The exponential weighting ensures that nearby molecules contribute more strongly to the index, while distant molecules contribute progressively less.

The time-averaged crowding index

$$\langle \eta_{\text{CI}}^{(i)} \rangle = \frac{1}{T} \sum_{t=1}^T \eta_{\text{CI}}^{(i)}(t) \quad (7)$$

provides a measure of the local environment over the simulation trajectory involving T number of time steps. Time averaging reduces the effects of short-range fluctuations and provides a stable measure of the persistent crowding environment experienced by the molecule.

The crowding index was used to characterize the local structural environment of TEMPO radicals within the SAM during molecular dynamics simulations. The temporal evolution of the index was evaluated along trajectories of representative TEMPO radicals located in the central region of the monolayer or near its edges. Simulations were performed at 300 K and 100 K to examine the influence of thermal fluctuations on the local packing environment.

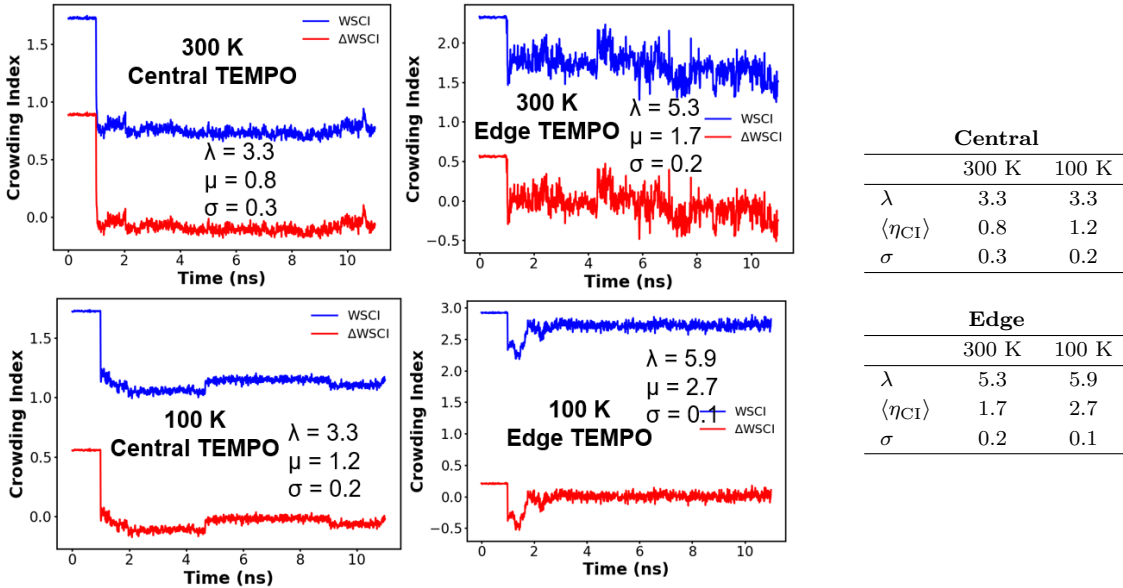


Figure 10: Spin crowding analysis (Equation 7) of the central and edge P-TEMPO at 300 K and 100 K, together with the corresponding weighted spatial crowding index (η_{CI}) parameters.

The trajectories reveal an initial relaxation stage that occurs at approximately 1 ns, during which the crowding index decreases rapidly. This behaviour reflects structural equilibration of the monolayer as the system relaxes from the initially constructed configuration toward a more energetically favourable packing arrangement. Following this equilibration period, the WSCI fluctuates around a stable mean value that characterizes the local environment of the radical. Clear differences are also observed between radicals located in the central region of the SAM and those near the edges. Molecules positioned in the centre experience a more symmetric and homogeneous packing environment due to the presence of neighbouring

molecules on all sides. This leads to relatively small structural fluctuations and a more stable local configuration. In contrast, TEMPO radicals near the edges of the monolayer experience a less uniform packing arrangement because fewer neighbouring molecules constrain their positions. As a result, the local structure is more dynamic, and the crowding index exhibits greater temporal variation. The higher effective crowding values and enhanced fluctuations indicate a more heterogeneous local environment in the edge regions.

These structural differences have important consequences for the magnetic environment of the nitroxide radical. Reduced fluctuations in the central region correspond to a more stable local magnetic field, which minimizes time-dependent perturbations that contribute to spin dephasing. Consequently, radicals located in the centre of the monolayer are expected to exhibit longer spin coherence times. In contrast, the enhanced structural variability at the edges introduces stronger environmental noise that can accelerate spin decoherence.

4.2.2 Conformational Dynamics of the Self-Assembled Monolayer

To characterize the dominant molecular motions in the TEMPO SAM, principal component analysis (PCA) was applied to molecular dynamics trajectories at 300 K and 100 K. The fractional contribution of each principal component is calculated by $\lambda_i / \sum_j \lambda_j$, where λ_i is the eigenvalue of the i^{th} principal component.

The first two principal components, defined as PC1 and PC2 capture a significant fraction of the total variance at both temperatures. At 300 K, PC1 and PC2 account for 40.4% and 19.4% of the variance, respectively, whereas at 100 K the contributions change to 46.4% and 17.4%. In each case, PC1 and PC2 together contribute $\sim 60\%$ of the total contribution arising from the eigenvalues. The larger contribution of PC1 at lower temperature indicates that molecular motion becomes more constrained and increasingly dominated by a single collective mode.

The distributions of key structural parameters further illustrate the effect of temperature on the internal geometry and flexibility of the TEMPO molecules. The N–O bond length distributions remain centred around ~ 1.30 Å at both temperatures, with only minor differences between molecules located in the central and edge regions of the SAM. This indicates that the primary bond structure of the nitroxide group remains largely preserved and is weakly affected by temperature. Similarly, the O–N–R₁ bond angle shows a narrow distribution centred near ~ 114 – 116° , again with minimal differences between central and edge molecules. These results suggest that the local bonding geometry of the nitroxide radical remains structurally stable within the monolayer environment.

More pronounced differences appear in the nitrogen pyramidalization and M1–R1–N–O dihedral angle distributions. At 300 K these coordinates exhibit broader distributions, partic-

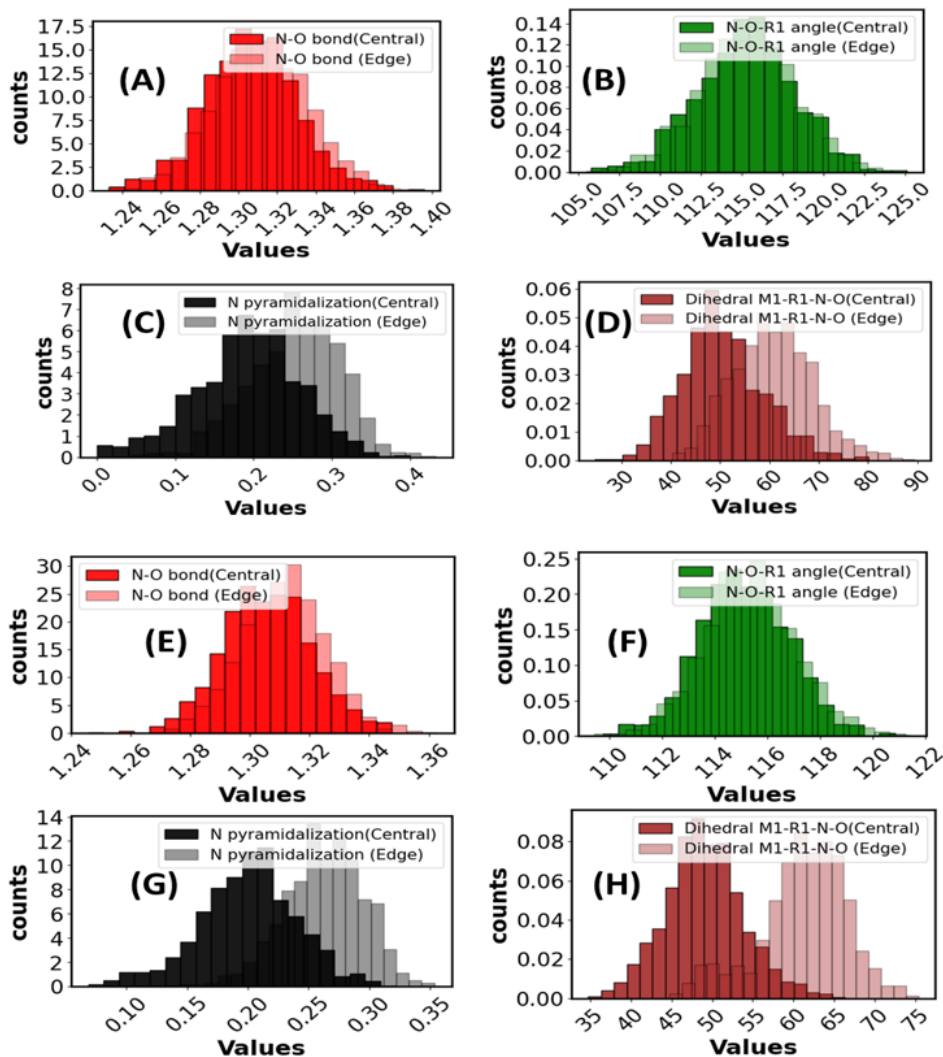


Figure 11: Internal coordinates resolution for P-TEMPO as mentioned (A)-(D): Temperature 300 K, (E)-(H): Temperature 100 K.

ularly for edge molecules, indicating enhanced conformational flexibility and larger torsional fluctuations due to weaker packing constraints. In contrast, at 100 K the distributions become significantly narrower, reflecting reduced thermal motion and increased structural rigidity. Nevertheless, edge molecules still display slightly broader distributions compared to central molecules, suggesting that reduced intermolecular constraints at the monolayer boundary allow greater local flexibility. Higher temperature promotes broader configurational sampling and enhanced torsional motion, whereas lower temperature restricts the system to fewer dominant vibrational modes and more localized structural fluctuations.

The free-energy landscapes (FEL) projected onto the PC1 and PC2 reveal the configurational sampling of the TEMPO-SAM system at different temperatures. At 300 K, the

landscape shows a broader basin along the PC1–PC2 coordinates, indicating that the system samples a wider range of conformations due to enhanced thermal fluctuations. On the other hand, at 100 K, the landscape becomes more localized with discrete low-energy regions, reflecting restricted configurational sampling and reduced thermal motion. The appearance of isolated minima suggests that the system is confined to a limited number of metastable states. Overall, decreasing temperature leads to a more confined free-energy landscape, demonstrating that thermal fluctuations play a key role in governing the dynamical flexibility of the system.

4.2.3 Spin-Spin Coupling in the Self-Assembled Monolayer

Spin–spin coupling between two radical centres can be described through the magnetic dipole–dipole interaction between their unpaired electron spins. This interaction depends on the magnetic dipole moments of the radicals and their spatial separation. Since the dipolar interaction scales as r^{-3} , the coupling is highly sensitive to the distance and relative orientation between radicals.

The spin–spin interactions between TEMPO radicals were analyzed by selecting a central TEMPO unit and evaluating pairwise magnetic exchange coupling constants (J) with neighbouring radicals at varying intermolecular separations. The exchange coupling constant J_{AB} was obtained from high-spin (HS) and broken-symmetry (BS) energies as [36],

$$J_{AB} = -\frac{E_{\text{HS}} - E_{\text{BS}}}{\langle S^2 \rangle_{\text{HS}} - \langle S^2 \rangle_{\text{BS}}}. \quad (8)$$

All distances reported here are measured with respect to this central reference with another radical (O–O distance), in the respective shells, and the angle reported is the dihedral angle between a pair of N–O of the corresponding P-TEMPO, allowing a direct assessment of how magnetic interaction evolves with spatial separation and relative orientation.

Table 2: O–O distance relative to a central unit and O–N–N–O dihedral dependent spin–spin interaction parameters for a pair of P-TEMPO radicals.

Distance (Å)	Angle (°)	J (cm ^{−1})
3.56	-22.45	-1263.47
4.84	105.13	-0.23
5.66	5.64	0.01

A clear distance dependence of the magnetic interaction is observed. The closest pair at 3.56 Å exhibits a large negative exchange coupling ($J = -1263.47$ cm^{−1}), indicating a strong preference for antiparallel spin alignment. Such a large magnitude suggests that at

short separations, the interaction is dominated by significant exchange contributions arising from effective interaction between the nitroxide spin centres. The relative orientation of the N–O groups, reflected in the angular parameter, further supports that specific geometric arrangements can strongly stabilise this antiferromagnetic coupling.

As the distance increases to 4.84 Å, the magnitude of J decreases drastically to -0.23 cm^{-1} , indicating that the interaction becomes extremely weak. At 5.66 Å, the coupling is effectively negligible ($J \approx 0$), suggesting that beyond a certain separation, the radicals behave as nearly independent spin centres. This rapid attenuation of J highlights the short-ranged nature of exchange interactions in the system. Importantly, the variation in J can not be attributed to distance alone. While the decay with separation is evident, the sign and magnitude are also influenced by the relative orientation of the interacting TEMPO units. Small changes in angular arrangement lead to significant differences in magnetic behaviour, indicating that both geometric and electronic factors govern the coupling.

Spin–spin interaction in the TEMPO assembly is highly localized, with strong coupling confined to very short intermolecular distances and rapidly diminishing beyond this range. Such distance-sensitive interactions are particularly relevant in the context of molecular spin systems, where control over the spatial arrangement directly influences coherence and interactions between spin centres.

4.3 Limitations of the Study

Spin-related properties are evaluated for static, minimum-energy geometries, while real systems undergo continuous thermal motion. Properties such as exchange coupling, g -tensor, and hyperfine interactions are highly sensitive to intermolecular distance and orientation, making single-point results configuration-dependent. The environment is also treated in a simplified way, limiting the description of collective effects, long-range interactions, and dynamic disorder.

5 Conclusions and Future Outlook

This work explores how molecular structure, vibrational motion, and spin relaxation are connected in TEMPO-based SAMs, with the goal of understanding their potential as molecular qubits. One of the main observations is the clear difference between central and edge TEMPO molecules within the SAM. The central molecules exist in a more ordered, constrained environment, with reduced crowding and smaller fluctuations, resulting in greater structural stability. In contrast, edge molecules are more exposed and therefore exhibit increased disorder.

der and larger structural variations. This difference directly impacts spin relaxation: central molecules experience lower noise and maintain coherence for longer, while edge molecules contribute more strongly to decoherence.

The vibrational analysis shows that not all modes contribute equally to spin-phonon coupling. Instead, a small number of low-frequency modes dominate the process, acting as efficient channels for energy exchange between the spin system and its surroundings. A comparison between TEMPO and P-TEMPO further highlights how chemical modification reshapes the vibrational landscape and redistributes coupling strengths, offering a pathway to relaxation.

Structurally speaking, the N-O bond remains relatively stable across different environments, preserving the radical character of TEMPO. However, fluctuations in angular and dihedral coordinates—particularly those involving out-of-plane distortions introduce significant changes in the local electronic environment. These variations affect parameters such as the g -tensor and hyperfine interactions, thereby contributing to quantum noise. In particular, out-of-plane motions play a dominant role in decoherence, while in-plane rigidity helps preserve coherence. The enhanced stability and reduced noise observed in central TEMPO molecules suggest that highly ordered SAMs are favourable for qubit applications. In contrast, edge effects and structural disorder must be minimized to achieve scalable and reliable quantum devices. Identifying the specific vibrational modes responsible for relaxation also opens the possibility of engineering systems where these modes are suppressed or shifted.

Several promising directions emerge from this study. A deeper, mode-resolved understanding of spin-phonon coupling could enable the predictive design of molecules with improved coherence properties. Additionally, optimization of surface interactions and linker chemistry could further enhance structural ordering and reduce environmental noise. The concept of ordered two-dimensional TEMPO arrays also suggests a pathway toward scalable qubit architectures. TEMPO SAMs provide a compelling example of how molecular design, surface chemistry, and quantum physics can be integrated to achieve this goal. The insights gained here contribute to a broader vision in which quantum devices are engineered at the molecular level, paving the way for advances in quantum computation, sensing, and information processing.

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A Some Sample Input Files

A.1 ORCA input file for AIMD

The following input was used for ab initio molecular dynamics simulations at 300 K:

```
!MD B3LYP DEF2-TZVP DEF2/J
%pal nprocs 40 end
%maxcore 2000
%md
  initvel 300_K
  timestep 0.5_fs
  thermostat berendsen 300_K timecon 10.0_fs
  dump position stride 10 filename "aimd-300k-traj.xyz"
  run 20000
end
* xyz 0 2
  C      -4.777124      -1.328110      0.330359
  <Coordinates>
  ...
*
```

A.2 ORCA input file for distortion along normal mode

```
program tempo_mode_analysis
  implicit none
  integer, parameter :: natom = 35
  integer, parameter :: nmode = 3*natom - 6
  integer, parameter :: nblock = nmode / 3
  real, parameter :: pi = 3.141592653589793
  real, parameter :: c_cm = 2.99792458e10 ! speed of light (cm/s)
  real, parameter :: h_bar = 6.3507e-4 !amu.cm2.sec-1
  integer :: i, j, k, ii
  integer :: nm1, nm2, nm3
  integer :: mass(natom)
  real :: Q(nmode, natom, 3)
  real :: w(nmode), mu(nmode), fc(nmode), ir(nmode)
  real :: rq, rtot
  real :: rNO, rRing, rMetH, rMetC, rProp
```

```

real :: omega, denom, nume, dimless_preff
character(len=15) :: dum1
character(len=2)  :: dum2
character(len=70) :: dum
character(len=2)  :: sym(nmode)
open(unit=16, file='fr.dat1', status='old', action='read')
do i = 1, nblock
  read(16,*) nm1, nm2, nm3
  read(16,*) sym(nm1), sym(nm2), sym(nm3)
  read(16,*) dum1, dum2, w(nm1), w(nm2), w(nm3)
  read(16,*) dum1, dum1, dum2, mu(nm1), mu(nm2), mu(nm3)
  read(16,*) dum1, dum1, dum2, fc(nm1), fc(nm2), fc(nm3)
  read(16,*) dum1, dum1, dum2, ir(nm1), ir(nm2), ir(nm3)
  read(16,*) dum  ! blank line
  do ii = 1, natom
    read(16,*) j, mass(ii), &
      (Q(nm1,ii,k), k=1,3), &
      (Q(nm2,ii,k), k=1,3), &
      (Q(nm3,ii,k), k=1,3)
  end do
end do
close(16)
write(*,'(a)') &
'Mode  Freq(cm-1)  ForceC   IR      dimeless_pref(cm)  NO      Ring      Meth
MetC'
do i = 1, nmode
  rNO   = 0.0
  rRing = 0.0
  rMetH = 0.0
  rMetC = 0.0
  rtot  = 0.0
  rProp = 0.0
  do j = 1, natom
    rq = Q(i,j,1)**2 + Q(i,j,2)**2 + Q(i,j,3)**2
    rtot = rtot + rq
    ! ---- NO group (atoms 1112)
    if (j == 11 .or. j == 12) rNO = rNO + rq
    ! ---- Ring carbons

```

```

        if (j >= 1 .and. j <= 5) rRing = rRing + rq
        ! ---- Methyl carbons
        if (j == 13 .or. j == 17 .or. j == 21 .or. j == 25) rMetC = rMetC + rq
        if (j == 29 .or. j == 30 .or. j == 33 .or. j == 34) rProp = rProp + rq
        ! ---- Methyl hydrogens
        if ((j >= 14 .and. j <= 16) .or. &
            (j >= 18 .and. j <= 20) .or. &
            (j >= 22 .and. j <= 24) .or. &
            (j >= 26 .and. j <= 28)) rMetH = rMetH + rq

    end do
    if (rtot > 0.0) then
        rNO    = rNO    / rtot
        rRing  = rRing  / rtot
        rMetH  = rMetH  / rtot
        rMetC  = rMetC  / rtot
        rProp  = rProp  / rtot
    end if
    omega = 2.0 * pi * c_cm * w(i)
    denom = mu(i) * omega
    nume   = h_bar
    dimless_preff = sqrt(nume/denom)
    write(*,'(i4,1x,f10.1,1x,f8.3,1x,f7.3,1x,es14.5,5(1x,f7.4))') &
        i, w(i), fc(i), ir(i), dimless_preff, rNO, rRing, rMetH, rMetC, rProp
    end do
end program tempo_mode_analysis

```

A.3 ORCA input file for evaluation of g and A anisotropy

```

!UKS B2PLYP D3BJ EPR-II AUTOAUX TIGHTSCF
%maxcore 3000
%pal nprocs 40 end
* XYZFILE 0 2 <input coordinates file (xyz)>
%EPRNMR
    GTENSOR TRUE
    NUCLEI = ALL N {AISO, ADIP, AORB}
    ORI GIAO
END

```


A.4 ORCA input file for spin-spin coupling

Broken symmetry calculation performed on 2 TEMPO radicals (triplet) to obtain the spin-spin coupling parameter, J (in cm^{-1})

```
! UKS CAM-B3LYP def2-TZVP TightSCF RIJCOSX

%maxcore 2000
%pal nprocs 40 end

%scf
    Brokensym 1,1
end

* xyz 0 3
O      101.661003      157.067001      178.735992
O      101.661003      158.660995      181.001999
N      102.922997      156.843002      178.835007
N      102.922997      158.451996      181.132996
<Coordinates of 2 TEMPO radicals- broken symm calc for triplet>
<The spin centres to be written at the top>
*
```

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